
Auditing Population Uncertainty in Cryo-EM: Numerical Integration, Nuisance Laws, and Identifiability

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Development manuscript; not submitted.

Abstract

A detailed cryo-electron microscopy reconstruction does not by itself support calibrated uncertainty about density or conformational populations. Starting from conjugate-paired Fourier Gaussian reconstructions of three experimental stacks, we examine whether population inference is a defensible next target. Matched simulations separate information discarded by moment summaries from numerical error in orientation marginalization. Fixed template-bank statistics have 29–55 times the squared separation of the best tested moment contrasts, but are inaccurate marginal likelihood approximations. One continuous importance-sampling repair passes an unchanged numerical gate on all three stacks. A simulation-only posterior-draw diagnostic shows why agreement between banks still cannot certify absolute integrals. We derive the implicit known-pose population bias caused by state-dependent viewing laws and show that bounded scores cannot be exactly unbiased over arbitrary continuous state-specific amplitude intervals. A frozen materiality screen then fails on all three compact-region targets: maximum modeled population biases are .000214, .000122 and .003493, below .02. These results end that proposed method branch. Recorded acquisition metadata and corner spectra expose further gaps between the simulator and experimental inference. We provide reproducible diagnostics and explicit limitations, not a validated population-uncertainty method or a claim of experimentally calibrated coverage.

1. Introduction

Cryo-EM reconstruction combines noisy two-dimensional projections, uncertain orientations and imaging parameters, and potentially heterogeneous molecular states. Neural implicit fields and Gaussian representations can recover complex structures, but representation quality and uncertainty calibration are different properties. Fourier shell correlation (FSC), posterior dispersion, independent-particle prediction and coverage of a population fraction answer different questions. An uncertainty study must specify its target before selecting a baseline or interpreting agreement.

Our project began with Gaussian kernels in Fourier space and subsequently studied confidence intervals for fixed density features. The resulting pose-aware intervals were uninformative in the tested end-to-end simulator; moment-based candidate tests were sensitive to uncalibrated viewing and noise assumptions. Rather than adding another interval to that construction, this manuscript asks a narrower question: *do the available data and numerical tools justify developing population uncertainty for two nominated structures?* The answer for the compact perturbations tested here is no.

This conclusion requires distinguishing three obstacles. A summary statistic may discard information even when its moments are computed exactly. A richer likelihood statistic may retain discrimination while failing as a numerical marginal likelihood. Finally, a proposed nuisance-robust population method may address an effect too small to motivate that method on its nominated targets. Improving the second obstacle does not resolve the third.

We contribute an auditable investigation of these obstacles, with fixed computational and scientific stopping rules. The numerical repair is classical importance sampling, not a new inference principle. The population-score identity and amplitude obstruction are elementary results with explicit scope; we do not claim priority for general mixture or analytic-continuation arguments. The empirical contribution is the matched three-stack diagnosis and its unfavorable decision, supported by saved arrays and independent calculations. An experimentally useful uncertainty method, a demonstrated advantage over matched population baselines, and submission-level novelty remain unresolved.

¹Fourier Cryo Splats research project. Correspondence to: Project repository <<https://github.com/yashpatel5400/fourier-cryo-splats>>.

2. Targets and Closest Prior Work

Reconstruction and density uncertainty. Bayesian cryo-EM reconstruction is established (Scheres, 2012). Gaussian pseudo-atom posterior inference (Joubert & Habeck, 2015), variational Fourier-slice uncertainty (Ullrich et al., 2020), and joint likelihood-Hessian analysis (Rangan et al., 2024) precede the present work. CryoDRGN and CryoDRGN-AI address heterogeneous representation learning and ab initio reconstruction (Zhong et al., 2021; Levy et al., 2025); a fixed-pose homogeneous fit does not reproduce their full tasks. Gaussian representations are likewise not new to cryo-EM (Chen & Ludtke, 2021; Chen & Ling, 2026; Qu et al., 2025; Shekarforoush et al., 2025).

Populations and likelihoods. BioEM marginalizes image nuisances and evaluates candidate structures (Cosio & Hummer, 2013); CryoLike supplies modern likelihood tooling (Tang et al., 2025). Cryo-BIFE already samples population/free-energy uncertainty along a nominated conformational path, using BioEM nuisance integration (Giraldo-Barreto et al., 2021). RECOVAR accounts for noisy latent coordinates in heterogeneity and population estimation (Gilles & Singer, 2025). Whole-stack ensemble reweighting can correct severe distortions from hard or soft particle counting (Evans et al., 2026). Thus posterior sampling or a two-state likelihood alone is not a contribution. The uncertainty target here would be a population *among selected particles*, not an equilibrium solution population unaffected by acquisition and selection.

Viewing laws and numerical marginalization. Nonuniform views matter in moment recovery (Sharon et al., 2020; Hoskins et al., 2026), structural comparison (Zhang et al., 2024a), and misspecified likelihood (Xu et al., 2025). The recent Bayesian orientation discussion explicitly considers estimating the view prior jointly with structure (Xu et al., 2026). Adaptive proposals, smoothing and a uniform defensive component were already used for cryo-EM marginalization by Brubaker et al. (2015). Our continuous quaternion proposal is numerical repair within that tradition. Agreement of independent approximations is weaker than a certificate of the integral; posterior-draw diagnostics for simulated data are also established (Grosse et al., 2015).

Validation and the calibration unit. Gold-standard halves and noise substitution address reconstruction overfitting (Scheres & Chen, 2012; Chen et al., 2013); confidence maps address signal significance (Beckers et al., 2019; 2020); independent-particle tests assess image fit (Ortiz et al., 2020). None is automatically a confidence interval for a biological population. CryoBench and CAHRA separate volume, latent-state, population and pose tasks (Jeon et al., 2025; Feathers et al., 2026). The broader survey in Ap-

pendix D documents related uncertainty targets and reading limits. Its 141 candidates are not 141 fully audited methods.

3. Forward Model and Gaussian Reconstructions

For $F(k) = \int \rho(x) e^{-2\pi i k^\top x} dx$, the observation model is

$$Y_i(q) = a_i C_i(q) e^{-2\pi i q^\top t_i} F_{Z_i}(E_i q) + \epsilon_i(q). \quad (1)$$

Z_i is molecular state, E_i the oriented central plane, t_i shift, C_i CTF and a_i real amplitude. Population inference must specify which of these are observed, marginalized or uncertain. Throughout our new numerical diagnostics, templates, one CTF and zero shifts are fixed; orientations are Haar and each realified Fourier-noise coordinate is independent $N(0, 1)$. These are simulator assumptions, not measurements of experimental nuisance distributions.

Our original reconstruction parameterization uses paired spectral kernels:

$$F_\theta(k) = \sum_j \{c_j G_j(k - \mu_j) + \bar{c}_j G_j(k + \mu_j)\}, \quad (2)$$

$$G_j(u) = \exp(-u^\top \Lambda_j u / 2), \quad \Lambda_j \succ 0.$$

Real precision matrices and conjugate pairing enforce $F_\theta(-k) = \overline{F_\theta(k)}$; self-conjugate terms are real. The dense differentiable reference allows movable centers and anisotropic precisions. The *actual three-stack benchmark* fits a sparse fixed lattice and isotropic width, with real/imaginary coefficients tied across conjugate nodes. It does not demonstrate adaptive anisotropic splatting. Tomographic projection is linear Fourier slicing, without optical alpha compositing.

The new diagnostics use the saved 64^3 fitted maps as constant-cell templates with exact cell Fourier factors, evaluated by NUFFT and selected independent direct sums. This makes off-grid orientation evaluation explicit; it does not preserve an unrestricted spectral representation or eliminate template error. Each map is normalized to unit array L^2 norm. A previously fixed Gaussian mask of width $20 \text{ \AA}$ is centered at its largest smoothed peak in the central half-field cube. Write $\rho_G = \rho M$ and $\rho_\delta = \rho - \delta \rho_G$. These are artificial local perturbations, not experimentally labeled conformations or measured biochemical occupancy.

4. Numerical and Statistical Diagnostics

4.1. Information retained by a fixed statistic

For a scalar score T , define

$$D(T) = \frac{(\mathbb{E}_1 T - \mathbb{E}_0 T)^2}{\text{Var}_0(T)}. \quad (3)$$

We compare power and power-plus-bispectrum contrasts, including separately trained classical covariance-aware directions, with a frozen template-bank log ratio. All use the same candidate, region, transfer, 220 frequencies and 8,192 paired noise/view draws. We retain both 32,768-point banks, four quadrature prefixes and all four nonzero deletion fractions.

A finite-bank log ratio remains a statistic if its quadrature is poor. Its measured D is therefore a descriptive separation, but it is not the KL divergence of the continuous image model or an information ceiling. We estimate the gap SE with the paired draws and the SE of D with a delta calculation that also includes variance-estimation uncertainty (Appendix C). No normal power guarantee is inferred.

4.2. A fixed orientation-integration repair

For a fixed image and map, integrate the positive Gaussian residual kernel $f_y(R)$ over normalized Haar measure, with integral I_y . For independent $R_l \sim q_y$,

$$\hat{I}_y = \frac{1}{L} \sum_{l=1}^L f_y(R_l)/q_y(R_l). \quad (4)$$

The omitted Gaussian normalization is common to the compared maps. The proposal is fit using the image and the midpoint template, then frozen before either importance bank. Attempt one combines eight locally optimized angular central Gaussian (ACG) quaternion components with .05 Haar mass. It fails the declared gate on two stacks.

The one permitted repair reuses all local fits and assigns masses .45 to the local mixture, .50 to a catalogue mixture and .05 to Haar. Catalogue centers are the same 32,768 training rotations; weights are proportional to the midpoint likelihood raised to $1/2$. Every center receives an isotropic ACG kernel with an 8-degree scale parameter, not an 8-degree angular standard deviation. Continuous rotations are sampled from this mixture, and its *full* density is evaluated using all centers. There is no top- k replacement of the target integral. Appendix B gives the density and all fixed choices.

Both attempts use the same 64 source indices, two generating states and two 8,192-draw banks per stack. A stack passes if at least 90% of its 128 images have bank log-ratio discrepancy at most .01, and all four median importance ESS values exceed 256. All three must pass. This is a necessary feasibility gate, not a sufficient numerical or inferential certificate.

4.3. Simulation-only posterior-draw check

The generating pose of a correctly simulated image is a posterior draw conditional on that image. This premise can expose a common missed mode even when two ordinary banks agree (Grosse et al., 2015).

Proposition 1 (Reciprocal identity). *Fix y and a positive integrable f_y with integral I_y . Let $X_1 \sim f_y/I_y$ and $X_2, \dots, X_L \sim q_y$ independently, with $q_y > 0$ on the support of f_y . For $I_\star = L^{-1} \sum_l f_y(X_l)/q_y(X_l)$, $\mathbb{E}[I_\star^{-1} | y] = I_y^{-1}$.*

The proof is exchangeability after changing the first draw's measure (Appendix A). We replace draw zero, not a selected weight, and use only the matching generating state. The premise does not hold for the wrong map on an image or for unknown experimental poses. For independent images, products of forward estimates and reciprocal planted estimates give pointwise stochastic log-integral brackets by Markov's inequality. Paired generating states are never multiplied together.

5. Population Bias and Amplitude Limitations

Let the selected-particle fraction of state A be $p \in (0, 1)$, and let states have view laws ν_A, ν_B . First suppose poses are observed and conditional image densities $f_A(y | R), f_B(y | R)$ are correct. A pooled conditional mixture uses $f_t = t f_A + (1 - t) f_B$, ignoring state-view dependence. Define

$$J_t(R) = \int \frac{(f_A - f_B)^2}{t f_A + (1 - t) f_B} dy. \quad (5)$$

Proposition 2 (Implicit pooled-population bias). *The expected pooled score is*

$$S(t) = p(1 - t) \mathbb{E}_{\nu_A} J_t - (1 - p) t \mathbb{E}_{\nu_B} J_t. \quad (6)$$

An interior maximizer $t_\star$ satisfies

$$\frac{t_\star}{1 - t_\star} = \frac{p}{1 - p} \frac{\mathbb{E}_{\nu_A} J_{t_\star}}{\mathbb{E}_{\nu_B} J_{t_\star}}. \quad (7)$$

Under strict concavity, $t_\star = p$ exactly when the expectations of J_p under the two view laws agree.

The odds equation is implicit because J_t depends on t . Different view laws alone do not imply bias. This is a known-pose mixture calculation, not a theorem for a latent-pose reconstruction pipeline.

For the following Gaussian calculations, images and means are realified by stacking real and imaginary coordinates; inner products use that real space. For unit covariance, $g = m_A - m_B$ and $s = \|g\|^2$, the log density ratio under B/A is $\sqrt{s}Z \mp s/2$. One-dimensional Gaussian quadrature therefore evaluates $J_t(s)$ exactly in this model, up to reported numerical error. The approximation $J_t(s) = s + O(s^2)$ yields a simpler odds factor, which we retain only as a secondary comparison to the exact root.

With known amplitude one and known poses,

$$\psi(Y, R) = \frac{\langle Y - m_B(R), g(R) \rangle}{\|g(R)\|^2} \quad (8)$$

has conditional mean equal to the state indicator. For any joint state-view law and independent unit-covariance noise,

$$\text{Var}(\bar{\psi}) = \{p(1-p) + \mathbb{E}_\nu[1/s(R)]\}/n, \quad (9)$$

provided the harmonic moment is finite, where $\nu = p\nu_A + (1-p)\nu_B$. View robustness therefore has a harmonic information cost in this restricted model.

For *real* amplitude, the local Gaussian-mean Schur complement at the full map is

$$s_\perp = \|g\|^2 - \frac{(\text{Re}\langle g, m_A \rangle)^2}{\|m_A\|^2}. \quad (10)$$

Replacing s by $s_\perp$ in Equation (9) is only a local feasibility surrogate. It does not prove that variance formula for a population estimator under arbitrary unknown amplitudes.

Proposition 3 (No exact bounded amplitude-unbiased score). *For fixed means m_A, m_B and nonsingular Gaussian noise, no bounded measurable score has conditional expectation 1 in state A and 0 in state B for every amplitude in a nonempty open interval for each state. The intervals may differ and need not contain zero.*

The Gaussian-smoothed expectation is entire in amplitude. Being constant on an interval forces the same constant at amplitude zero for each state, where the image laws coincide, a contradiction. The proof is given in Appendix A. This rules out exact zero-bias bounded feature calibration over those intervals; it does not rule out approximate bias-aware intervals, prove non-identifiability, or establish a convergence rate. Finite-catalogue zero residual cannot be interpreted as continuous nuisance validity.

6. Experiments and Decisions

6.1. Data and comparison scope

We use EMPIAR-10028, 10049 and 10076 (EMPIAR, 2015a;b; 2016). Each original reconstruction uses 8,192 experimental particles downsampled to 64^2 , with 7,374 fitting, 409 validation and 409 test particles. The two fitting halves share public consensus poses, so their FSC is conditional, not fully gold-standard. The sparse Gaussian and trilinear baselines have mean inter-method FSC .9920, .9806 and .9909 over the recorded sampled shells. These high values measure agreement, not truth or coverage.

Stock homogeneous neural training, voxel backprojection and unknown-pose RELION runs are retained separately. Registered-reference mean FSC for the three RELION results is .702, .202 and .122; the 10076 reference is a Class A assembly state, not heterogeneous-consensus truth. These are reconstruction comparisons, not uncertainty baselines. The exact protocols, curves and historical failures remain linked in Appendix E.

Table 1. Matched 25% deletion. Known-pose twice-KL is an oracle quantity; the other columns are squared separation D of the best tested moment and bank-0 template statistics.

EMPIAR	Oracle 2 KL	Moment D	Bank D
10028	0.05119	0.001617	0.04752
10049	0.09347	0.001735	0.06858
10076	0.01757	0.000122	0.00674

All new simulation and metadata comparisons below are development analyses. Their individual protocols were committed before their new outcomes, but they were motivated by earlier failures. The 65,536 training views and 8,192 paired image draws are reused explicitly; these are not new independent confirmations. All orientations, masks, seeds and unfavorable cases are retained. New numerical work ran on an M4 Pro Mac with 24 GB memory.

6.2. Matched discrimination and nuisance exposure

At 25% deletion, the frozen template statistic has about 29, 40 and 55 times the best tested moment separation (Table 1). The bank-0 D estimates have delta SEs .000764, .00120 and .000146. The second bank gives .047142, .068197 and .006741. These comparisons identify compression loss under the matched simulator without treating the finite-bank statistic as a converged marginal likelihood.

Moment means are also sensitive to nuisance misspecification. For matched power scores, relative white-noise variance errors of .514%, 6.86% and .263% match the 25%-deletion mean gaps. View-dependent amplitudes in $[-.9, 1.1]$ cover that mean gap on 10028 and 10076. These are mean-envelope diagnostics, not finite-sample test guarantees. The richer likelihood statistic has not yet been tested with experimentally calibrated colored noise, per-particle CTF variation or unknown amplitude. Comparing its ideal-model score to nuisance-stressed moments cannot establish superior experimental robustness.

6.3. Continuous integration: a computational pass

The original 32,768-point template banks have null ESS medians 1.23, 7.75 and 28.7. Their individual log ratios remain unstable despite similar mean separation. The first local continuous proposal places 128/128, 103/128 and 79/128 images within .01 between banks. Its final repair increases these to 128/128, 124/128 and 119/128, passing the unchanged all-stack gate (Table 2, Figure 1).

The four final median ESS values range from 2,245–2,255, 2,234–2,262 and 1,265–1,347. Lower-decile ranges are 2,131–2,138, 1,323–1,427 and 512–550. The three concurrent runs take 22.65–22.71 minutes each; this is not a matched speed comparison with matrix-cached template banks. All 384 integral summaries replay. Full-covariance

Table 2. Unchanged .01 ratio-discrepancy criterion. Every repaired stack also passes the median-ESS requirement.

EMPIAR	First	Repair	Repair RMS
10028	128/128	128/128	0.000887
10049	103/128	124/128	0.004519
10076	79/128	119/128	0.005074

density checks at 8,192 fixed points per stack agree within 1.6×10^{-13} in log density; 18 direct cell residual checks agree within 7×10^{-11} .

The planted-truth audit retains a failure of the bank-agreement heuristic: one old 10076 *within-tolerance* case changes by 1.50 log-integral units after replacing a single draw by truth. After repair, the largest such jump is .0626. The exact integrals remain unknown. Absolute log-integral bank RMS after repair is .0288, .0296 and .0424, larger than ratio RMS .000887, .00452 and .00507. No original local Hessian eigenvalue was clipped, so covariance clipping does not explain these discrepancies. The numerical branch ends after this single permitted repair.

6.4. Recorded nuisances: experimental assumptions remain open

The full source metadata contain 105,247, 108,544 and 131,899 particles. For 128 equal-area bins of unoriented plane normals, peak recorded-density ratios to uniform are 4.53, 6.23 and 2.74; cross-half chi-square summaries are 1.07, 2.01 and .248. They are inconsistent with treating these recorded histograms as near-Haar. Estimated poses, symmetries and selection prevent interpreting them as exact latent distributions or continuous density bounds. All 90 original histogram vectors independently reproduce.

Four 8×8 particle-corner patches per selected particle show nonflat DCT power, after accounting for Fourier cropping. The relative-coordinate power ranges are .366–2.101, .783–3.000 and .551–2.162. Particle tails, small windows and unknown stationarity preclude declaring these pure-noise spectra. Existing STAR/Frealign joins recover 229, 137 and 351 acquisition groups for the 8,192-particle cohorts. An earlier missing-group statement was wrong and is corrected with a complete join/split audit. Between-group fractions of patch-energy sums of squares are .159, .0724 and .117; these are descriptive ANOVA fractions, not an intraclass-correlation model.

Per-particle defocus varies substantially, and recorded scale/weight fields include constants that cannot calibrate physical amplitudes. The alpha-view association of 2.62% on 10028 and .74% on filtered 10076 is a fraction of recorded alpha variance associated with bins, *not* a percentage amplitude bound. A single-CTF, known-white-noise simulator remains an explicit idealization.

Table 3. Primary population screen: largest absolute bias and harmonic amplitude-tangent surrogate range across all prelisted assignments. The same pair must meet both criteria; all fail materiality.

EMPIAR	Max. $ t_\star - p $	Harmonic range	Gate
10028	0.000214	1.291–1.292	Fail
10049	0.000122	1.039–1.039	Fail
10076	0.003493	3.872–3.916	Fail

6.5. Population materiality: a scientific failure

We freeze A as the full fitted map and B as its *full* masked-region deletion, distinct from the 25%-deletion discrimination study. All 65,536 saved training rotations are replayed, with direct physical checks at the first, batch-boundary, halfway and final indices. At $p = .75$, the known-pose pooled root is evaluated for every prelisted pair: source halves on all three stacks, and published filtering versus its complement on 10028/10076, each in both state assignments.

These metadata groups are *not biological states*: all three downloaded class fields are constant zero. Each group’s recorded full-rotation histogram defines a piecewise-Haar proxy on a fixed ZYZ grid. The primary grid has 8^3 cells; 4^3 and 12^3 are sensitivity analyses. This is a controlled calculation under proxies, not evidence that two actual conformations have those laws.

A stack passes only if the same primary pair/assignment has $|t_\star - .75| \geq .02$ and $\mathbb{E}_\nu[1/s_\perp] \leq 8.8125$. The latter gives a nominal local-surrogate SE .03 at $n = 10^4$ after adding $.75 \times .25$; it is not an unknown-amplitude coverage theorem. At least two stacks must pass. No sensitivity grid, new region or relaxed threshold can rescue a failure.

All three fail materiality (Table 3); no sensitivity bias reaches .02 either. Every harmonic surrogate passes, so low modeled population bias, rather than poor local separation, ends this proposed method branch. Direct expected-score quadrature independently checks all 30 roots, with residual at most 3.1×10^{-11} . Histogram bins and projected energies reproduce independently. This does not prove general adequacy of pooled likelihood or impossibility of population inference.

6.6. Existing population baseline and external controls

We execute unchanged author functions from Evans et al. (2026) on all 23 released two-state likelihood conditions, retaining both successful and numerically discrepant results. A separate scalar optimizer verifies the pooled optimum; the weakest synthetic author result differs by .00785, with small float32 simplex drift. For the published experimental default, the independent estimate is .798228; under the condition labeled rotation-10 it is .649749. This is pose-

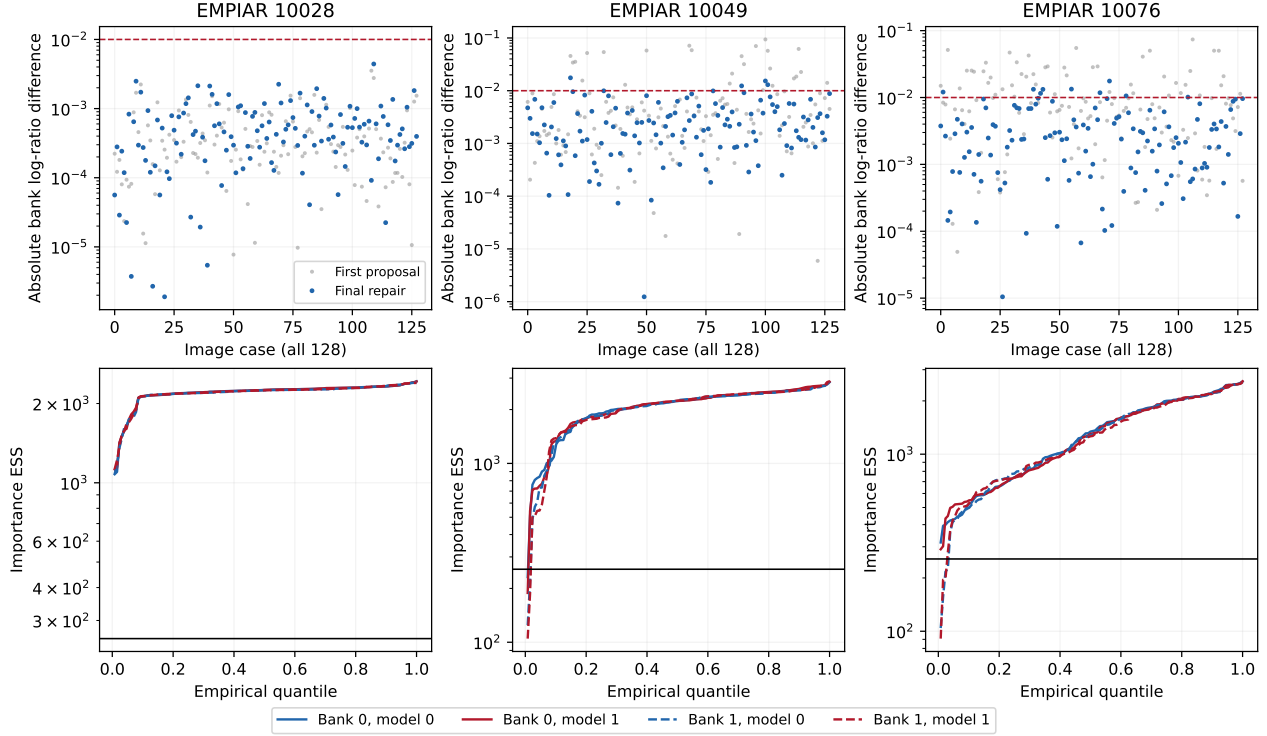


Figure 1. All 384 image cases of the final integration repair. Top: absolute bank log-ratio differences, retaining the first proposal in gray and the repair in blue; the threshold is .01. Bottom: each bank/model ESS distribution, with the median gate’s threshold 256 shown for reference. Thirteen repaired cases remain outside tolerance. Axis ranges differ by stack. Passing this necessary numerical gate does not establish an absolute-integral certificate or experimental uncertainty coverage.

model sensitivity, not a measured state-view population bias. The .8 experimental reference is a constructed mixture of classified subsets, not biological truth.

The official CAHRA pose-convention supplement provides 216 high-signal projections per state. All 432 signed quarter-turn comparisons are retained; the correct in-plane sign gives median correlation about .988, the opposite sign about .74. This checks a coordinate convention only. We have not evaluated the separate noisy 30,000-particle pose/population challenge or its experimental constructed-mixture task. The requested benchmark access, noise-only windows and biological state labels remain missing evidence.

7. Interpretation and Remaining Requirements

Three distinctions survive these experiments. A low-dimensional summary can lose discrimination without establishing a limit on the original data. A numerical integration gate can pass while absolute-integral error remains too large or insufficiently controlled for inference. A nuisance-robust population proposal can fail its materiality screen despite plausible local information. These are separate findings, not interchangeable success claims.

The exact known-pose odds identity diagnoses a specific misspecification; its extension to latent views and imperfect templates is not supplied. The amplitude proposition requires nonzero response bias to remain explicit in bounded-feature constructions. General bias-aware inference, mixture optimization and deconvolution are established foundations (Donoho, 1994; Armstrong & Kolesár, 2018; Lindsay, 1983; Zhang et al., 2024b; Pensky, 2017). No universal population-coverage theorem or new minimax rate is claimed.

Further method development needs an externally motivated state pair, a calibrated acquisition/noise model, matched whole-stack population baselines, repeated unconditional calibration and informative experimental controls. Repeated fits or model reviews cannot substitute for this evidence. Four full independent model reviews of earlier drafts rejected the work; the latest focused consultation motivated the bounded checks here and did not provide an acceptance verdict. This rewritten diagnostic manuscript has not received a fifth full review. The work remains an incomplete research program rather than a demonstrated ICML-level uncertainty method.

Reproducibility and Impact

All code, failed attempts, source hashes, numerical arrays and review records are public at <https://github.com/yashpatel5400/fourier-cryo-splats>. The v0.7.7/v0.7.8 evidence releases and their v0.7.6 dependencies provide the new experiments; source provenance distinguishes downloaded papers from executed author code. Gaussian fitting, metadata and conditional simulations must not be advertised as calibrated biochemical populations. The principal impact of this work is an auditable account of where the proposed inference does and does not have support.

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A. Proofs and Their Scope

These arguments support the stated diagnostic interpretations. They are not claimed as a new general framework for inference in mixtures or inverse problems. In particular, correct algebra conditional on a noise or nuisance model does not establish that model on experimental particles.

A.1. Pooled conditional-mixture identity

For $f_t = tf_A + (1-t)f_B$, write $u_t = (f_A - f_B)/f_t$. Since $\int (f_A - f_B) = 0$,

$$\int f_A u_t = \int \{f_t + (1-t)(f_A - f_B)\} u_t \quad (11)$$

$$= (1-t)J_t,$$

$$\int f_B u_t = -tJ_t. \quad (12)$$

Taking expectations over the *state-specific* laws gives Equation (6). An interior zero of that score gives Equation (7). The second derivative of a conditional log likelihood is $-u_t^2$, so its expectation is nonpositive, and strictly negative whenever the conditional components differ with positive probability. Under the stated strict-concavity condition the zero is the unique interior optimum. At $t = p$ its score is $p(1-p)(\mathbb{E}_{\nu_A} J_p - \mathbb{E}_{\nu_B} J_p)$, proving the final statement. One must evaluate J at the fitted fraction; replacing it by a fixed discriminability ratio changes this identity into an approximation.

For Gaussian means, let $\ell = \log(f_A/f_B)$ and $h_t(\ell) = \text{logistic}(\text{logit}(t) + \ell)$. Then $u_t = (h_t - t)/(t(1-t))$. Subtracting its two conditional expectations, or using the preceding identities, yields For $Z \sim N(0, 1)$, this gives

$$J_t(s) = \frac{\mathbb{E}\{h_t(\ell_+) - h_t(\ell_-)\}}{t(1-t)}, \quad (13)$$

$$\ell_{\pm} = \sqrt{s}Z \pm s/2.$$

The implementation integrates the difference before averaging, reducing cancellation for small s . Roots use 64-node Gauss–Hermite quadrature on $[10^{-7}, 1-10^{-7}]$ with root tolerance 10^{-10} ; 128 nodes check the normalized score against 10^{-6} . A separate verifier integrates the direct density-ratio score under both generating Gaussians using a distinct 160-node quadrature implementation. None of these checks certifies a latent-pose marginal likelihood.

A.2. Harmonic variance and the amplitude tangent

Conditionally on state $Z \in \{0, 1\}$ and pose R , the score in Equation (8) is $Z + \langle \epsilon, g(R) \rangle / s(R)$. Its conditional mean is Z and its conditional variance is $1/s(R)$. Total variance gives $p(1-p) + \mathbb{E}_{\nu}[1/s(R)]$; independent images divide that variance by n . No state–view independence is used.

The argument requires correct means, amplitude one, conditional zero-mean unit-covariance noise, $s > 0$ and a finite harmonic moment. It neither proves unbiasedness after noisy pose estimation nor permits arbitrary view-dependent amplitudes.

At a Gaussian mean with perturbation direction g and real-amplitude direction m_A , the Fisher Gram matrix is

$$\begin{pmatrix} \|g\|^2 & \text{Re}\langle g, m_A \rangle \\ \text{Re}\langle g, m_A \rangle & \|m_A\|^2 \end{pmatrix}. \quad (14)$$

Its Schur complement is Equation (10). This is a *mean-parameter* calculation, not the efficient information of a binary mixture with unrestricted amplitude laws. The secondary projection at m_B is retained in every screen array. Under Haar, projection at the full map reduces arithmetic mean separation by 4.95%, 37.14% and 8.28%. The corresponding harmonic surrogates are 1.2912, 1.1206 and 3.9184. Nonzero local separation is compatible with nonlinear ambiguity or unstable population inference.

A.3. Bounded-score amplitude obstruction

Whiten a nonsingular Gaussian covariance. For bounded measurable ψ , set $F_z(a) = \int \psi(y) \phi_d(y - am_z) dy$. For complex $a = u + iv$, the modulus of the extended Gaussian integrand is bounded by

$$\|\psi\|_{\infty} \phi_d(y - um_z) \exp(v^2 \|m_z\|^2 / 2). \quad (15)$$

On compact subsets of a , this and the derivatives needed to differentiate under the integral have integrable Gaussian tails. Hence F_z is entire. If $F_A = 1$ on a nonempty real interval, the identity theorem gives $F_A = 1$ everywhere. Likewise $F_B = 0$ on its interval implies $F_B = 0$ everywhere. But $F_A(0) = F_B(0) = \mathbb{E}\psi(\epsilon)$, a contradiction.

Zero amplitude is used for analytic continuation, not assumed physically admissible. A fixed affine combination of finitely many bounded features is bounded, so the proposition applies to any finite-coefficient calibrated feature score. It does not supply a quantitative bias–variance lower bound, rule out nonlinear multi-image estimation or establish a root- n impossibility result. General linear-functional deconvolution theory is relevant (Pensky, 2017), but its detailed rate results have not been verified for our multivariate, support-constrained model.

A.4. Reciprocal importance identity and stochastic brackets

For positive f and q , set $w = f/q$, and let $I = \int f$. Changing the first measure from f/I to q gives

$$\begin{aligned} \mathbb{E}[I_{\star}^{-1}] &= I^{-1} \mathbb{E}_{q^L} \left[\frac{Lw(X_1)}{\sum_l w(X_l)} \right] \\ &= I^{-1}. \end{aligned} \quad (16)$$

The last step uses exchangeability: the L expectations of $w(X_j)/\sum_l w(X_l)$ are equal and sum to one. Positivity ensures the denominator is nonzero. A fixed replacement index is valid; choosing a replacement according to observed weights is not this experiment.

For a matched joint simulation, generating latent variables conditional on the observations have the exact posterior law. This familiar identity underlies bidirectional Monte Carlo (Grosse et al., 2015). The proposal may depend on the image and a fixed independent catalogue, but it must not use the true pose beyond information already in the image. Other draws must be conditionally independent of that posterior point. Conditional on an already fixed generating pose, the posterior-draw interpretation is absent.

For independent images, $\prod_i \hat{I}_i$ is unbiased for $\prod_i I_i$, and $\prod_i I_{\star,i}^{-1}$ is unbiased for its reciprocal. Markov and a union bound give

$$\sum_i \log \hat{I}_i - \log(1/\delta_L) \leq \sum_i \log I_i, \quad (17)$$

$$\sum_i \log I_i \leq \sum_i \log I_{\star,i} + \log(1/\delta_U) \quad (18)$$

jointly with probability at least $1 - \delta_L - \delta_U$. We use $\delta_L = \delta_U = .025$ for each fixed 64-image state/bank bracket. There is no simultaneous claim across stacks, states or banks; paired states share noise and poses and are never multiplied together. The four final widths per stack range from 7.395–7.398, 7.404–7.415 and 7.508–7.519 log units. The additive floor $2 \log 40 = 7.378$ dominates. These are stochastic simulation brackets, not a practical uniform numerical certificate for experimental inference.

A.5. Identifiability is distinct from an unbiased score

For completeness, consider fixed templates with compact noiseless mean supports $\mathcal{M}_0, \mathcal{M}_1$ generated by a compact nuisance space and continuous forward map. Allow arbitrary joint state/nuisance laws and assume measurable probability lifts from each mean support to its nuisance space. The latter is automatic for a finite catalogue; it is an explicit assumption here for continuous spaces.

Let μ be the law of the noiseless observed mean, and add independent known nonsingular Gaussian noise. Its characteristic function is nowhere zero, so Gaussian convolution identifies μ from the infinite-data observed distribution. The compatible state-1 fractions are exactly

$$[\mu(\mathcal{M}_1 \setminus \mathcal{M}_0), \mu(\mathcal{M}_1)]. \quad (19)$$

Only state 1 can generate the first set; all state-1 mass lies in the second. Conversely, assign any desired fraction of intersection mass to state 1 and lift the resulting measures. Thus disjoint supports identify the population, despite the

bounded-score obstruction above. This is a classical support/deconvolution argument, not evidence of stable finite-data recovery or of separation between the actual continuous cryo-EM supports.

An admissible pair of means a, b , with whitened distance d , gives a simple finite-sample obstruction. The all-state-0 and all-state-1 Gaussian experiments have $\text{KL} = nd^2/2$, hence $\text{TV} \leq \min(1, \sqrt{nd}/2)$. Any interval C covering the population at level $1 - \alpha$ in both experiments satisfies under the first

$$\Pr(\{0, 1\} \subset C) \geq \max\{0, 1 - 2\alpha - \min(1, \sqrt{nd}/2)\}. \quad (20)$$

Transfer the second coverage event by total variation and use a union bound. A found, independently checked close pair would be sufficient for this negative witness; a failed local search would not certify positive separation. We do not report such a numerical witness in this study. Density floors or restricted nuisance laws can invalidate point-mass witnesses and must not be silently discarded.

B. Numerical Protocol and Independent Checks

B.1. Proposal density on rotations

The ACG construction normalizes a four-dimensional Gaussian vector to a unit quaternion (Tyler, 1987). For covariance $\Sigma \succ 0$, its density relative to normalized uniform measure on the unit sphere is $|\Sigma|^{-1/2}(u^\top \Sigma^{-1} u)^{-2}$. It is antipodally symmetric, so the same normalized density passes to rotations. This expression follows by integrating the Gaussian radial coordinate; we do not claim a new distribution or a full audit of Tyler’s paper.

For catalogue center c and $s = (8 \text{ degrees in radians})/2$, use $\Sigma_c = s^2 I + (1 - s^2)cc^\top$. The component density is

$$s^{-3}\{[1 - (c^\top u)^2]/s^2 + (c^\top u)^2\}^{-2}. \quad (21)$$

The local proposal uses eight optimized midpoint modes. Its three-dimensional Hessian covariance is divided by four in the quaternion tangent coordinates; the scalar quaternion coordinate has unit variance. The original precision clamps correspond to .25–30 degree standard deviations. No saved optimum’s eigenvalues actually hit these clamps. Original failed optimizer statuses are retained, and reused modes are not counted as newly converged fits.

The target maps in the integration experiment are ρ_0 and $\rho_{.25}$; their midpoint is $\rho_{.125}$. Catalogue weights are proportional to $\exp(\text{midpoint log-kernel}/2)$. Full mixture density is evaluated over all 32,768 centers in chunks of 256 points. The catalogue broadens the proposal; the integrand remains a continuous-orientation physical likelihood. Haar mass .05 provides support and finite variance for these bounded

kernels, but does not ensure useful variance at this draw count.

Each stack uses integer-linspace indices from 0 to 8,191, with 64 indices and both generating states. Bank seeds are $261020 + \text{dataset} + 100 \times \text{position} + 10 \times \text{state} + \text{bank}$. Both 4,096-point prefixes and 8,192-point endpoints are retained. The repair changes the sampling algorithm, not selected seeds. Maps, observations, draw counts and gate thresholds stay fixed.

B.2. Replay, samples and diagnostic errors

All saved log integrals, ESS values, maximum weights, ratio differences and gate decisions are recomputed from kernels and proposal densities. The independent density implementation uses inverse full 4×4 covariances at 32 fixed integer-linspace draws from each bank/image: 8,192 independently checked points per stack. Physical checks explicitly sum all 64^3 cells at all 220 frequencies for six selected orientations per stack, evaluating both maps. These are floating-point consistency checks, not outward-rounded error certificates or exhaustive direct-sum checks.

For normalized importance weights w_{lm} under model m , the plug-in delta variance for a shared-draw log ratio is $L/(L-1) \sum_l (w_{l1} - w_{l0})^2$. The between-bank SE combines their variances. Heavy or missed tails can make this estimate unreliable. We retain matching-state weight mass on Haar-labeled draws, the source family of the largest weight, Hessian clipping, and the fixed-index planted jump for every case. Stratification by the original .01 gate is explicitly post hoc, not a predictive validation.

B.3. Rotation regeneration and law proxies

The population screen regenerates all 65,536 training rotations using seed 261017 + dataset. Each batch draws 512 rotations, then the original 512×220 real and imaginary noise arrays. Skipping those intervening draws would produce different later rotations. Direct cell checks at indices 0, 511, 512, 32,767, 32,768 and 65,535 verify both full and region means; maximum discrepancy is 1.08×10^{-11} .

Full-rotation bins use intrinsic ZYZ coordinates $\alpha/(2\pi), (\cos \beta + 1)/2, \gamma/(2\pi)$, with periodic angular wrapping. Equal-width cells have equal Haar volume. For metadata mass p_c and n_c training rotations in cell c , each such rotation receives weight p_c/n_c . An occupied metadata cell with no training draw would make that grid unavailable; no such case occurred. The independent verifier derives Euler angles from rotation-matrix entries and reconstructs every histogram. These finite histograms define the tested proxies. They are not an estimate with guaranteed error against the true continuous state-specific viewing law.

C. Sampling Errors and Complete Outcome Scope

Let $x_i = T_{0i}$, $h_i = T_{1i} - T_{0i}$, $\hat{\mu} = \bar{h}$ and $\hat{v} = (n-1)^{-1} \sum_i (x_i - \bar{x})^2$. A delta influence for $\hat{D} = \hat{\mu}^2/\hat{v}$ is

$$\hat{\varphi}_i = \frac{2\hat{\mu}}{\hat{v}}(h_i - \hat{\mu}) - \frac{\hat{\mu}^2}{\hat{v}^2} \left\{ \frac{n}{n-1}(x_i - \bar{x})^2 - \hat{v} \right\}. \quad (22)$$

We report $\text{sd}(\hat{\varphi}_i)/\sqrt{n}$ as an asymptotic conditional Monte Carlo SE. It includes gap/variance covariance; the SE of the gap alone is not an SE of D . Three empirical-contamination finite differences verify this influence for every one of 96 entries. No finite-sample confidence bound or power guarantee is attached to it.

Population-law linear summaries report within-cell Monte Carlo SEs conditional on fixed bin counts: if v_c is the sample within-cell variance, the estimated variance is $\sum_c p_c^2 v_c / n_c$. This does not include pose, template, histogram-estimation or noise-model error. The exact root is checked numerically, not assigned an experimental confidence interval. Both assignments of each law pair and all three grids are retained; no outcome-guided subgroup or cell selection is performed.

The study's accounting units are important. There are 384 integration image cases, formed from 192 source images across three stacks and two paired states; the paired states are not independent source particles. Each attempt has two banks per case. There are 30 population pair/assignment/grid comparisons, not 30 independent biological experiments. The template-score ledger has 96 rows sharing images, models and banks. Method comparisons use shared draws intentionally; row counts must not be treated as replication counts for statistical significance.

D. Critical Survey and Unresolved Scientific Questions

The living survey is dated 1 October 2026. It contains 141 curated bibliographic candidates and a much larger broad-search ledger that is not an individually read methods corpus. Source manifests record versions, access failures and reading depth. We distinguish published papers from versioned preprints, targeted full-text reading from abstract screening, and original-code execution from descriptions of published experiments. This is a substantial scoped survey, not a claim of exhaustive field coverage.

Density posteriors and model error. RELION's Bayesian formulation, Gaussian pseudo-atom inference and probabilistic Fourier-slice models are direct uncertainty antecedents (Scheres, 2012; Joubert & Habeck, 2015; Ullrich et al., 2020). Their posterior assumptions should

not be erased by calling them deterministic baselines. Likelihood Hessian tails expose weak density–pose directions (Rangan et al., 2024). Dynamic empirical-Bayes/MCMC approaches also study confidence and resampling targets (Lai et al., 2023). An additional posterior ensemble is therefore not novel merely because its representation is Gaussian. Meaningful comparison requires equal nuisance assumptions, information and scale, including a matched prior baseline where appropriate.

Pose distributions and confidence statements. Posterior-averaged rotations (Xu et al., 2026) and probabilistic search in SIMPLE (Van et al., 2025) address estimation. CryoForum’s dispersion supports ranking and filtering (Chung, 2023). CESPED explicitly uses refinement-derived experimental labels with their own uncertainty (Sanchez-Garcia et al., 2024); ARCHER and cryoPARES investigate transfer and pose prediction (Nguyen & Pham, 2026; Sanchez-Garcia et al., 2026). Their scores cannot be substituted for a uniformly valid per-particle error radius without checking the guarantee and deployment relationship. The recent Bayesian orientation paper’s Appendix F discusses joint prior estimation and sensitivity; simply fitting a preferred-view prior is already an explicit direction.

Heterogeneity representations. CryoDRGN, DynaMight and 3DFlex describe different density or motion families (Zhong et al., 2021; Schwab et al., 2024; Punjani & Fleet, 2023). RECOVER uses covariance estimation and kernel regression (Gilles & Singer, 2025). Physical heterogeneity is a distribution of molecular states, not automatically posterior uncertainty about a single mean density. Independent-half deformation disagreements and held-out motion validation have specific targets; shared learned components and evaluated frequency bands matter. Gaussian GMMs, CryoSplat, GEM and hierarchical parts are substantial representation prior art (Chen & Ludtke, 2021; Chen & Ling, 2026; Qu et al., 2025; Shekarforoush et al., 2025). Fourier conjugate pairing differs from real-space splats, but needs demonstrated value beyond that distinction.

Population and thermodynamic uncertainty. CryoBIFE integrates image nuisances and samples free-energy profiles on a fixed path (Giraldo-Barreto et al., 2021). Our follow-up reads the main synthetic sections, all seven supplement pages and all six supplemental figures; we do not execute its MCMC. Its controls vary noise, defocus, count and path, and include posterior-versus-MAP/assignment comparisons. A visible reference outside an interval in one realization is not a frequentist coverage study. The method’s nuisance integration must not be replaced by a weak known-pose baseline when comparing future work.

Whole-stack population inversion and classification counts

can differ substantially (Evans et al., 2026); our 23-condition author-code replay supports that computational distinction using supplied likelihoods. Measurement-limited ensemble granularity (Mattingly et al., 2026) and discretization error (Mordant et al., 2026) directly constrain claims about resolvable populations. cryoTWIN targets thermodynamic landscapes (Yamazaki et al., 2026). Targeted SI-3/SI-4 reading finds training-subset/seed and fixed-landscape path variability measures, rather than repeated-sampling coverage. Its comparisons include resolution and density-construction differences that require matching in our own studies. We did not audit all its proofs, figures or code.

Population uncertainty versus selection. Even accurately estimating the fraction among retained images does not recover solution-state populations without assumptions about particle capture, picking, classification and filtering. Constructed mixtures provide controlled source proportions but not complete molecular density truth. Selection and pose-model sensitivity must not be conflated with state-view dependence. The author-released rotation-10 result in our replay demonstrates sensitivity to a changed likelihood input, not its causal attribution to a biological viewing law.

Moment and distributional methods. Nonuniform moment reconstruction, compressed moment recovery and moment-based candidate comparison already estimate or profile viewing laws (Sharon et al., 2020; Hoskins et al., 2026; Zhang et al., 2024a). Their bandlimit, genericity, identifiability and acquisition assumptions matter. High-noise likelihood theory (Fan et al., 2024) does not calibrate individual poses. Recent moment-conditioned posterior sampling and functional deconvolution study distinct compressed-information models (Janson & Andén, 2026; Al-Ghaffas et al., 2026); their full code and proofs were not replicated here. Distributional inverse stability (Sanchez Espinosa et al., 2026) further distinguishes image-distribution fit from latent reconstruction error under explicit stability assumptions. The failure of our low-order scores cannot rule out collective recovery.

Map validation and learned priors. Independent halves, noise substitution, directional resolution and independent-particle validation have complementary roles (Scheres & Chen, 2012; Chen et al., 2013; Vilas et al., 2020; Ortiz et al., 2020). FDR confidence maps test density significance (Beckers et al., 2019; 2020). Blush regularizes reconstruction using learned information (Kimanus et al., 2024); LocScale-2.0 attaches confidence to map enhancement against its specified proxy (Bharadwaj et al., 2026). Bias/variance discussions already explain why consensus is insufficient (Sorzano et al., 2022). Sharper appearance, half-map agreement and enhancement confidence do not share one calibration claim, and cannot be ranked by an

inappropriate single coverage metric.

Atomic and multimodal targets. EMMIVox combines molecular information with map-based likelihoods (Hoff et al., 2024); experiment-guided AlphaFold3 addresses structural ensembles with multiple measurement types (Maddipatla et al., 2026). CalPro’s coordinate-error target uses a different calibration unit (Shihab et al., 2026). An atomic-model score is neither a raw-particle density interval nor a population confidence bound. Independent modalities can test interpretations that an image model alone leaves ambiguous.

Conformal and simultaneous calibration. Posterior-variance calibration and quantile/conformal inverse imaging rely on a specified population of signal–observation pairs (Narnhofer et al., 2024; Ye et al., 2026). Noise draws or neighboring voxels from one molecule are not automatically exchangeable new molecules. Simultaneous inverse confidence regions (Battle et al., 2025) differ from many separately calibrated marginal intervals. Bias-aware optimal recovery and GP/ridge connections are also established (Donoho, 1994; Armstrong & Kolesár, 2018; Kanagawa et al., 2025). A conditional theorem, an empirical error ranking and marginal population coverage require different evidence.

Gaussian uncertainty beyond cryo-EM. Bayesian splatting, photometric uncertainty, view-structured conformal prediction and radiative CT consider different forward models and targets (Jia et al., 2026; Galappaththige et al., 2026; Chu et al., 2026; Zhao et al., 2026). Their existence further limits a claim based only on attaching variance to Gaussian kernels. Translating a method requires checking the change from view synthesis or CT measurements to latent-pose Fourier projections, and from prediction error to structural or population inference.

Benchmark construction and remaining access. CAHRA version 2 describes three separate challenges (Feathers et al., 2026): experimental compositional mixtures, simulated conformational variation, and pose/population entanglement. The last uses two CALHM2 structures with preferred views and Gaussian noise, sampling state and view independently. A 3% uniform component is a density *floor*, not an upper bound on peak viewing density. Public benchmark proportions are known, so subsequent use is not a blind challenge entry. The separate convention supplement has been downloaded and checked here; the main 8 GB pose benchmark remains behind a Globus sign-in requirement. Its live leaderboard is not a substitute for a methodological comparison with identified baselines and uncertainty metrics. Challenge-1 frame overlap and common preprocessing must be checked

before treating frame sums as independent measurements.

Open questions supported by this survey. The strongest unresolved questions are population uncertainty under template and viewing-law error; calibrated absolute pose confidence; validation of detail introduced by learned priors; and propagation through CTF estimation, preprocessing, alignment and molecular interpretation. These are conclusions about priorities, not claims that no relevant prior work exists. For this project, a real source-labeled mixture and a defensible noise model are more valuable than another minor variant of the failed compact-region method.

E. Historical Comparisons and Reproduction

The original reconstruction and uncertainty studies are not erased by this rewrite. Their PDFs, source snapshots and exact outcomes remain in tagged releases. The new manuscript reorganizes the scientific question; it does not convert old conditional results into experimental calibration.

Reconstruction baselines. Three 8,192-particle datasets have Gaussian, trilinear and stock homogeneous neural fits using public consensus poses. The Gaussian/voxel test NMSE pairs are .83313/.83939, .93023/.93103 and .87503/.87909. These are normalized image prediction errors in the original preprocessing, not population error. Conditional half-map FSC remains above its threshold at the sampled limit, so nominal resolutions 16.08, 7.872 and 13.973 Å are censored limits, not independently resolved atomic detail. Backprojection is named separately from neural training. Unknown-pose RELION finishes on all three stacks with its convergence and registered-reference diagnostics retained. No ab initio CryoDRGN-AI result is substituted for those distinct experiments.

Density-interval failures. The earlier continuous audit compared a finite fitting dictionary with a larger supported density class. A matched continuous Gaussian-prior comparator shares its fixed-pose guarantee at the stated scale, so coverage was not an advantage over that baseline. The complete matched rerun has 48 converged fits and 672 conditional cases. A separate local-refinement study has 600 simulated datasets and 12,000 weight fits; all 216 reported procedure/control cells cover broad targets in 200/200 trials, but all three pose-audit procedures select the no-data fallback. Raw median widths are 642–1,242 times that fallback’s width. A local optimizer worsens the near-truth initialization; it is not a production-refinement comparison. Noise-only coverage of one pilot-aligned 10049 feature is .720 versus .955 with true poses in the saved-draw replay. These findings motivate caution about plug-in alignment but do not prove a collective cryo-EM limit.

Moment methods and closed development.

Power/bispectrum variants, classical concentration comparators, covariance-aware contrasts and equal-noise-budget allocations remain in the prior evidence releases. Their scope is controlled candidate validation, not population coverage. The new matched ledger and metadata audit address the information and nuisance questions those studies left open. No new moment variant is selected in this manuscript. The historical allocation text contained a transcription .000182; its source result is approximately .0001755. This does not affect the new matched comparison or gate outcomes.

Author-code population replay. The upstream commit is `b9099e1b7fb3`; the full identifier is retained in the replay manifest. Its original likelihoods are reused, not regenerated. All 10 synthetic and 13 experimental conditions are retained; two separately named noisy-image inputs are byte-identical and are not independent controls. The synthetic empirical label fraction .80074 differs from generator probability .8. The original float32 weights deviate from sum one by up to 1.82×10^{-5} ; raw and normalized diagnostics remain visible. The replay uses current isolated dependencies, not an exact reproduction of the authors' environment. It executes their functions without copying their implementation into our package.

Executable provenance. Separate runners, verifiers and reporters implement catalogue integration, population materiality, the saved-draw integration autopsy and fixed-bank separation. Exact script names and commands are in the repository's [post-review reproduction document](#). Archive manifests enumerate source commits, input hashes, byte lengths and dependencies. Replays refuse to overwrite their original records; a new audit needs a separately named output. We retain failed dependency/packaging attempts and correct their provenance rather than deleting them.

Independent review and resource use. The exact external model was `claude-fable-5-1`; four full earlier reviews reject at confidence 4/5. The post-round-4 methods consultation read supplied text, did not execute arrays or browse papers, and suggested the bounded population screen. Its unchanged visible response, prompt and provider identity are preserved; private reasoning is excluded. The catalogue repair was frozen and started before its critique was retrieved, although the provider had completed it shortly earlier. The actual .45 local mass does not follow its later local-majority suggestion. No parameters were changed to produce a favorable result. A model's approval would not establish conference acceptance, and no fifth full review has been requested for this diagnostic rewrite. All current work uses the Mac; no GPU rental is needed for the completed diagnostics.